# MAS508 Control Theory

**Project Description**

# State observer and feedback control of electro-mechanical system for lifting mass payloads with elasticity and disturbances

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# 1 Introduction

In various industries such as offshore, maritime, logistics, construction and production lines, among others, controlled mechatronic systems often operate dynamic mechanical loads. One very common practical application is an automatic lifting by the systems of ropes and cables, as seen in Figure 1. Despite the



Figure 1: Example of a maritime lifting system.

overall complexity of such systems, one of the main control challenges (in vertical direction) is due to the elasticity between the heavy mechanical payloads and the actuators driving the cable system(s). Even when restricting the dynamics analysis to only one vertical degree of freedom, multiple factors should be taken into account for designing a robust, safe, and reliable control system. Among those are:

a) Placement of the actuating and sensing elements;

b) Uncertain or unknown parameters of the payload, such as mass and mass distribution;

c) Elasticity in cables, ropes, joints and possible oscillatory behavior.

d) Environment disturbances;

e) Measurement accuracy, noise, and access to internal system states.

# 2 System Description

The electro-mechanical system considered in this project, shown in Figure 2, is a scaled system designed to replicate a typical application of operating dynamic mechanical loads with elasticity.

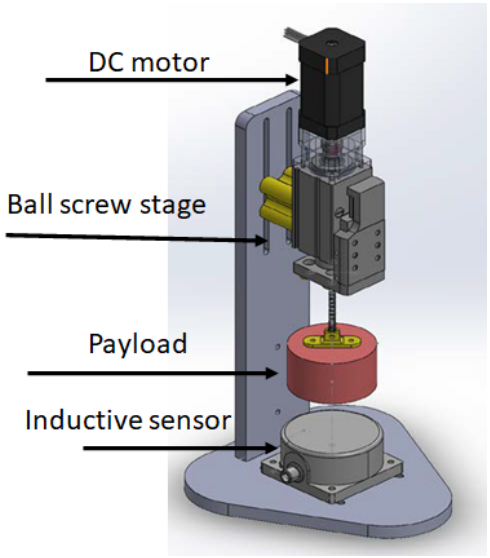



Figure 2: Project Setup

The system is actuated by a **ball-screw mechanism** driven by a **Brushless DC (BLDC) motor**. This actuation system is capable of lifting and lowering a mechanical payload with a constant mass $m_l$. The payload is connected to the actuator through a **spring** with elastic stiffness $k_s$. The spring elongates under the influence of gravity and simultaneously induces vertical oscillations when excited by the actuator's motion.

The objective of this project is to **model the given system** and subsequently **design a control system** that enables rapid lifting and lowering of the payload while minimizing oscillations.

An **inductive sensor** is mounted at the bottom to measure the payload position $x(t)$, as illustrated in Figure 3. The actuator has a **maximum stroke of 25 mm**, and its position is denoted by $y(t)$, although it is not directly measurable. The symbol $x_0$ represents the equilibrium position of the payload, corresponding to the spring's fully elongated state under gravitational force.

The system parameters and variables are given in Table 1 and Table 2. Figure 3 shows the physical configuration and the associated quantities, from which the state variables are to be defined.

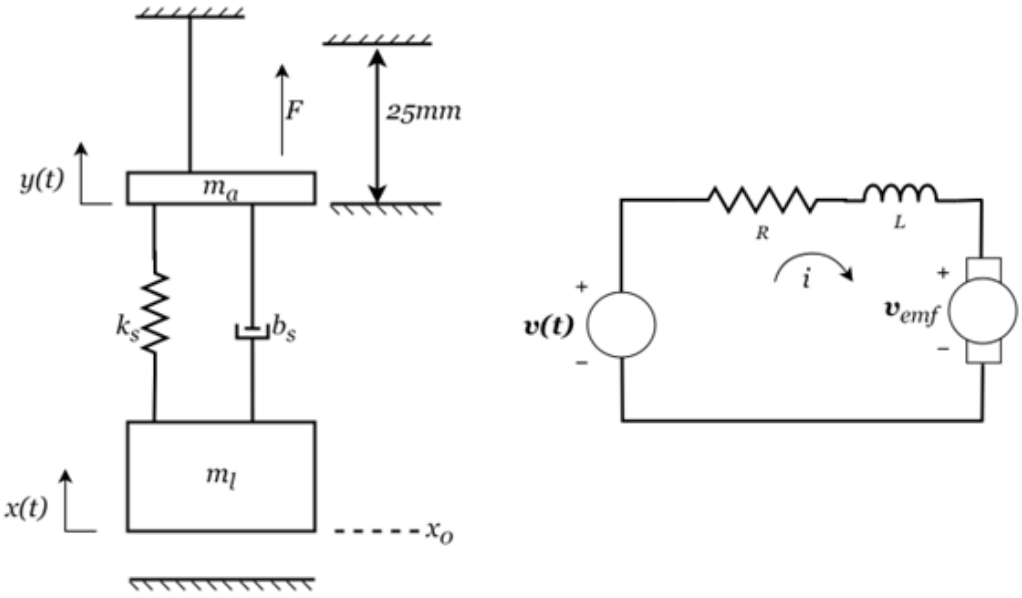



Figure 3: System diagram

| Parameter | | Value | Units |
|---|---|---|---|
| $m_a$ | Moving mass of actuator | 0.3 | kg |
| $R$ | Resistance (motor) | 0.55 | Ω |
| $L$ | Inductance (motor) | 0.8 | mH |
| $k_i$ | Torque constant | 0.036 | Nm/A |
| $k_b$ | Back emf constant | 0.036 | V/rad/s |
| $J_m$ | Motor Inertia | $72 \times 10^{-6}$ | $\mathrm{kg\,m^2}$ |
| $k_s$ | Spring coefficient | – | N/m |
| $b_s$ | Damping coefficient | – | Ns/m |
| $l$ | Pitch/lead of ball screw | 0.01 | m |
| $\eta$ | Ball screw efficiency | 1.0 | – |
| $m_l$ | Payload mass | 0.51 | kg |

Table 1: System parameters

| | | Range | Units |
|---|---|---|---|
| $y$ | Actuator position | [0 0.025] | m |
| $F$ | Actuator force | | N |
| $i$ | Armature current | | A |
| $v$ | Input (control) voltage | [-5 5] | V |
| $v_{emf}$ | Back emf | | V |
| $x$ | Payload position | [0 0.05] | m |

Table 2: System variables

# 3 Project Tasks

## 3.1 Modeling and simulation

- A **mathematical model** of the system must be developed. The absolute position of the payload, measured by the inductive distance sensor, is the only available state measurement.
- A **control-oriented model** of the system must be formulated, either as a block diagram, a set of differential equations, or directly in state-space form. The **unknown spring and damping coefficients** must be determined experimentally.
- The developed and analyzed system model must be **identified and validated** using real-time (RT) data generated and recorded from the experimental setup. The identification results should be evaluated in the frequency domain, time domain, or a combination of both.
- The **ball-screw stage** is used to convert rotary motion into linear motion. Only the axial motion of the ball screw is to be considered (see Appendix for details). The ball screw efficiency is assumed to be ideal $\eta = 1$, but the real effeciency is below 1 and is not known exactly; it is part of the model uncertainty that the controller must robust against.
- A **control system** must be designed, implemented, and tested in simulation to achieve fast lifting and lowering of the payload with minimal oscillations. The designed controller should later be verified on the real-time experimental setup.

## 3.2 Real-Time (RT) implementation

All mathematical models of the plant and control system must be developed in MATLAB/Simulink, analyzed, implemented, and tested in real time (RT), and properly documented as part of the project. The Speedgoat RT platform will be used to operate the physical setup. Its input/output channels will be utilized to send control commands and acquire position feedback. The designed control system must be implemented and validated in real time using the physical setup.

## 3.3 Control Specifications

1. A **full-state feedback controller** shall be designed to fulfill the following objectives:
   (a) as short a settling time as possible,
   (b) minimal transient and steady-state oscillations,
   (c) high tracking accuracy.
2. The unmeasured dynamic states of the system must be estimated through the design and implementation of a **state observer**.
3. The desired motion for the application is to lift the payload to $x_l = x_0 + 15\,\text{mm}$ and lower it to $x_l = x_0 + 5\,\text{mm}$. Additional reference positions of $x_l = x_0 + 20\,\text{mm}$ and $x_l = x_0 + 5\,\text{mm}$ should also be tested using the same controller, without any redesign or re-tuning. These are the minimum required test cases; additional test scenarios may be included for bonus credit.
4. **Steady-state accuracy** of the payload's output position must be ensured by the appropriate design of the overall control system.

5. The complete control system must be designed to be as **robust** as possible, accounting for modeling and identification uncertainties, as well as large force disturbances like step force, or a push.

### 3.4 Design Choices and Justification

The choice of identification method, controller design approach and observer design approach is part of the task. All choices must be propeorly justified.

## 4 Resources

1. There are two physical setups, numbered 1 and 2, placed in machine hall (Mezzanine Floor). Each setup is equipped with a Speedgoat. Every group will use its designated setup. The details will be posted on canvas.

2. Matlab 2025b will be used for RT testing. Speedgoat library for Matlab 2025b is available on canvas. ("speedgoat_io_blockset_9_11_0_R2025b_build_41981.zip").

   - Install Matlab2025b.
   - Install **Simulink Real-Time Target Support Package** from **Add-Ons**
   - Install Speedgoat library: Unzip the library and navigate to root folder of library and run **speedgoat_setup**.
   - The Speedgoat target has the IP address `192.168.7.1`. Use the `ping` command to verify that the connection is established. Do not change the IP address of speed gaot.
   - Open `SLRT Explorer` to connect to the Speedgoat target.
   - If the connection is not established, temporarily disable the Windows Firewall and try again.

3. A Matlab script "ctrb_calc.m" to check controllability via Hautus controllability test is also available on canvas. Use this script to confirm results from standard Matlab controllability functions.

4. A Simulink file **(Project_IOinterface.slx)** to interface I/O with the speedgoat is available on canvas. This file configures the inputs and outputs as shown in Figure 4.

5. Change the springs if deformed. Additional springs can be collected from my office.

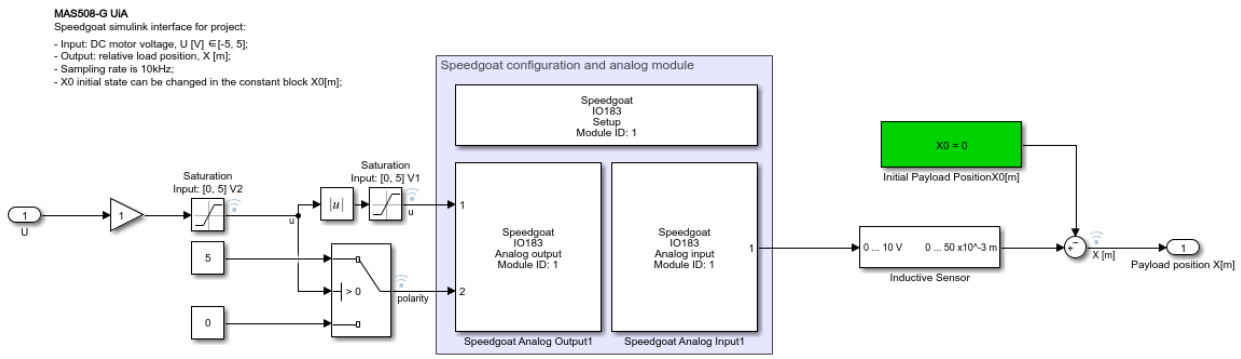



Figure 4: Simulink Interface

## 5 Submission and Assessment

1. Students must organize in groups of three students using Canvas. One submission per group must be delivered, which should contain:

(a) Project Report: should reflect working steps, approaches, justifications and results achieved. The written report and presentation should indicate the names of the students working together in one project group.

(b) MATLAB/Simulink model: files used for implementing and simulating the model and controller should be included.

(c) A video file with movie recorded from experiments on the real system must be created and submitted together with the project report

2. A mandatory project presentation and oral exam in groups will be done during week 47/48 (date to be announced later).

3. The project report and the additional simulation files should be submitted in Canvas. The deadline for submission is 10th November 2026 at 12:00.

4. Project demonstration will be carried out on 10th November 2026 at 12:30. Schedule will be posted on canvas.

5. The project will be assessed based on three components: the submitted project report, the project presentation, and the oral examination.

   The oral examination will be used to clarify technical details and assess the group's understanding of the project, including the methods, design choices, implementation, and results. The group should be prepared to explain and justify its technical decisions and demonstrate a thorough understanding of the work presented.

   The oral examination will have a significant impact on the group's final grade.

## 6 Use of AI Tools

Students may use AI tools to support their learning, clarify concepts, explore alternative solutions, and assist with project development. However, the submitted report and presentation must reflect the group's own work, critical thinking, and understanding.
A report that is wholly AI-generated and does not demonstrate meaningful contributions, critical evaluation, or understanding by the group may be rejected, subject to the applicable institutional regulations.
During the oral examination, the group must be able to explain and justify the technical concepts, design decisions, implementation, and results presented in its report. An inability to demonstrate an adequate understanding of the submitted work may result in a significant deduction in the group's final grade.
**Using AI is not prohibited in itself. The group remains responsible for the accuracy, quality, originality, and understanding of all submitted work.**

## 7 Plagiarism

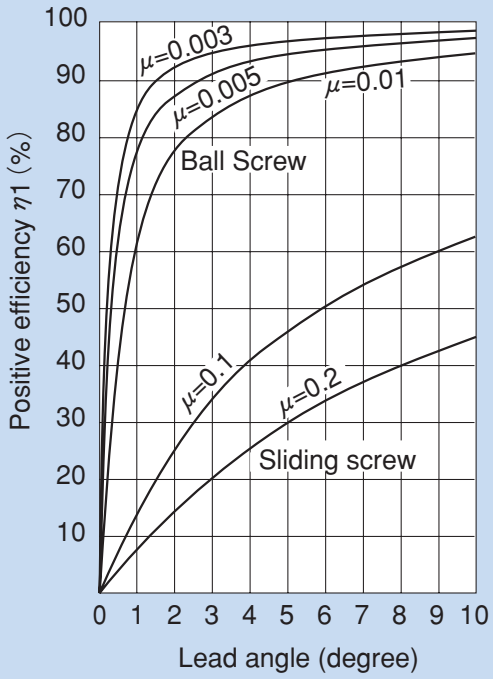
Students are expected to complete their projects independently. Using or copying material from previous years' projects is not allowed. We have access to all past submissions, and any instances of plagiarism will be addressed in accordance with university guidelines.

# 8 Appendix

# 1. Features of the Ball Screw

## 1.1. Driving Torque One Third of the Sliding Screw

With the Ball Screw, balls roll between the screw shaft and the nut to achieve high efficiency. Its required driving torque is only one third of the conventional sliding screw (Fig. 1 and 2). As a result, it is capable of not only converting rotational motion to linear motion, but also easily converting linear motion to rotational motion.



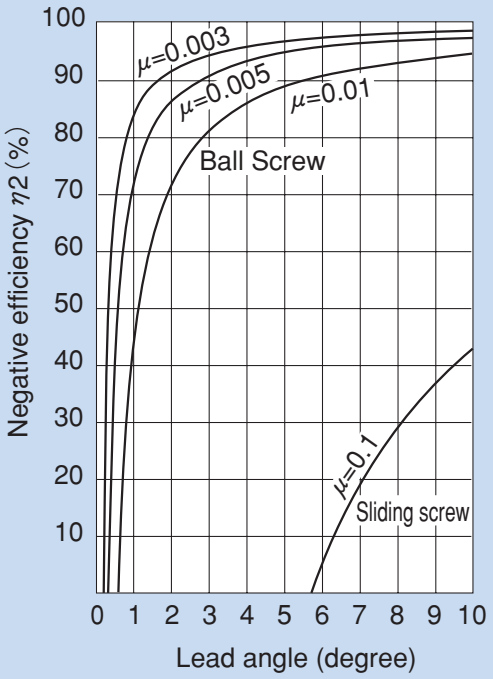




Fig. 1 Positive Efficiency (Rotational to Linear) Fig. 2 Negative Efficiency (Linear to Rotational)

### 1.1.1. Calculating the Lead Angle

$$\tan\beta = \frac{\ell}{\pi \cdot d_P} \quad \cdots\cdots\cdots(1)$$

where

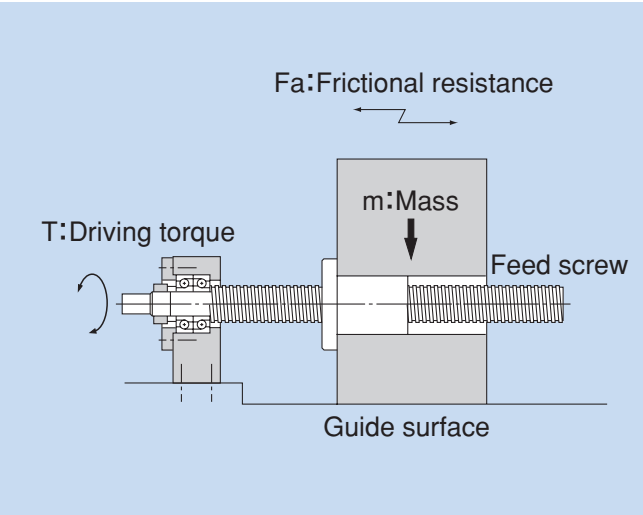
| | | |
|---|---|---|
| $\beta$ | :Lead angle | (degree) |
| $d_P$ | :Ball center diameter | (mm) |
| $\ell$ | :Feed screw lead | (mm) |

# 1.1.2. Relationship between Thrust and Torque

The torque or thrust generated when thrust or torque is applied is obtained from equations (2) to (4).

## Driving Torque Required to Gain Thrust

$$T = \frac{Fa \cdot \ell}{2\pi \cdot \eta 1} \quad \cdots\cdots\cdots (2)$$




where

T : Driving torque (N·mm)
Fa : Frictional resistance on the guide surface (N)
Fa = $\mu$ × mg
$\mu$ : Friction coefficient of the guide surface
g : Gravitational acceleration (9.8m/s²)
m : Mass of the transferred object (kg)
$\ell$ : Lead of the feed screw (mm)
$\eta$1 : Positive efficiency of feed screw
(Fig. 1 on page K-5)

## Thrust Generated When Torque is Applied

$$Fa = \frac{2\pi \cdot \eta 1 \cdot T}{\ell} \quad \cdots\cdots\cdots (3)$$

where

Fa : Thrust generated (N)
T : Driving torque (N·mm)
$\ell$ : Lead of the feed screw (mm)
$\eta$1 : Positive efficiency of feed screw
(Fig. 1 on page K-5)

## Torque Generated When Thrust is Applied

$$T = \frac{\ell \cdot \eta 2 \cdot Fa}{2\pi} \quad \cdots\cdots\cdots (4)$$

where

T : Torque generated (N·mm)
Fa : Thrust input (N)
$\ell$ : Lead of the feed screw (mm)
$\eta$2 : Negative efficiency of feed screw
(Fig. 2 on page K-5)